

Customer Shopping Behavior Analysis

End-to-End Data Analytics Project • Python • SQL • Power BI

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SECTION 01 Project Overview

This project presents a comprehensive end-to-end analysis of customer shopping behavior using transactional data covering **3,900 purchases** across multiple product categories. The objective is to uncover actionable insights into spending patterns, customer segments, product preferences, discount impact, and subscription behavior — enabling data-driven strategic business decisions.

■ Python	■ SQL (MySQL)	■ Power BI
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SECTION 02 Dataset Summary

3,900 Total Records	18 Features / Columns	37 Missing Values (Review Rating)
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Feature Group	Columns Included
Customer Demographics	Age, Gender, Location, Subscription Status
Purchase Details	Item Purchased, Category, Purchase Amount (USD), Season, Size, Color
Shopping Behaviour	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

SECTION 03 Exploratory Data Analysis using Python

■ Data Loading

Imported the raw dataset using `pandas`. Verified shape, dtypes, and initial row samples.

■ Initial Exploration

Used `df.info()` and `df.describe()` to understand data distribution, identify numeric vs categorical columns, and flag outliers.

■ **Missing Data Handling**

Detected **37 null values** in the `review_rating` column. Imputed using the **median rating per product category** to preserve distribution without data leakage.

■ **Column Standardisation**

Renamed all columns to **snake_case** for consistency and easier SQL integration.

■ **Feature Engineering**

Created `age_group` by binning customer ages into Young Adult / Middle-aged / Adult / Senior. Created `purchase_frequency_days` from frequency text field.

■ **Data Consistency Check**

Verified redundancy between `discount_applied` and `promo_code_used`. Dropped `promo_code_used` as it was fully correlated.

■ **Database Integration**

Connected Python to **MySQL** using SQLAlchemy and loaded the cleaned DataFrame directly into the database for downstream SQL analysis.

SECTION
04

Data Analysis using SQL — Business Transactions

Ten structured business questions were answered using MySQL, covering revenue analysis, customer segmentation, product performance, and discount behaviour.

#	Business Question	Key Finding
Q1	Revenue by Gender	Male: \$157,890 Female: \$75,191
Q2	High-Spending Discount Users	839 customers used discounts yet spent above average (~\$60)
Q3	Top 5 Products by Avg Rating	Gloves (3.86) › Sandals (3.84) › Boots (3.82) › Hat (3.80) › Skirt (3.78)
Q4	Express vs Standard Shipping	Express avg: \$60.48 Standard avg: \$58.46
Q5	Subscriber vs Non-Subscriber Spend	Subscribers: avg \$59.49 Non-subscribers: avg \$59.87 (similar)
Q6	Most Discount-Dependent Products	Hat (50%) › Sneakers (49.66%) › Coat (49.07%)
Q7	Customer Segmentation	Loyal: 3,116 Returning: 701 New: 83
Q8	Top 3 Products per Category	Clothing: Blouse, Pants, Shirt Footwear: Sandals, Shoes, Sneakers
Q9	Repeat Buyers & Subscriptions	Of >5 purchases: 2,518 not subscribed vs 958 subscribed
Q10	Revenue by Age Group	Young Adult: \$62,143 (highest) › Senior: \$55,763 (lowest)

SECTION
05

Dashboard in Power BI

An interactive Power BI dashboard was built to visualise key findings. It includes slicers for Subscription Status, Gender, Category, and Shipping Type, allowing stakeholders to drill down dynamically.



Fig 1 — Customer Shopping Behavior Power BI Dashboard

- **KPI Cards:** Number of Customers (3.9K) · Avg Purchase Amount (\$59.76) · Avg Review Rating (3.75)
- **Donut Chart:** Subscription Status breakdown — Yes 27% vs No 73%
- **Bar Charts:** Revenue by Category and Sales by Category — Clothing leads both
- **Horizontal Bars:** Sales and Revenue by Age Groups — Young Adults drive the highest values
- **Interactive Slicers:** Filter by Subscription Status, Gender, Category, and Shipping Type

SECTION 06 Business Recommendations

Based on the analysis across Python EDA, SQL queries, and Power BI visualisation, the following strategic recommendations are proposed:

1. Boost Subscription Rates

Only 27% of customers are subscribed despite similar spend levels. Introduce exclusive subscriber perks (early access, free shipping) to drive conversion and improve lifetime value.

2. Customer Loyalty Programme

80% of customers are already in the 'Loyal' segment (>10 purchases). Formalise a tiered loyalty reward scheme to reduce churn and increase basket size.

3. Optimise Discount Strategy

Products like Hat (50% discount rate) and Sneakers (~50%) show heavy discount dependency. Review margin impact and test lower discount thresholds to protect profitability.

4. Target Young Adult Segment

Young Adults generate the highest revenue (\$62,143). Allocate marketing budget toward digital channels and categories favoured by this cohort.

5. Product Positioning

Highlight top-rated items — Gloves, Sandals, and Boots — in campaigns. Promote Clothing category (highest revenue) with cross-sell opportunities.

6. Express Shipping Promotion

Express shipping users spend slightly more on average (\$60.48). Offer targeted promotions to Standard shipping users to upsell them to Express.